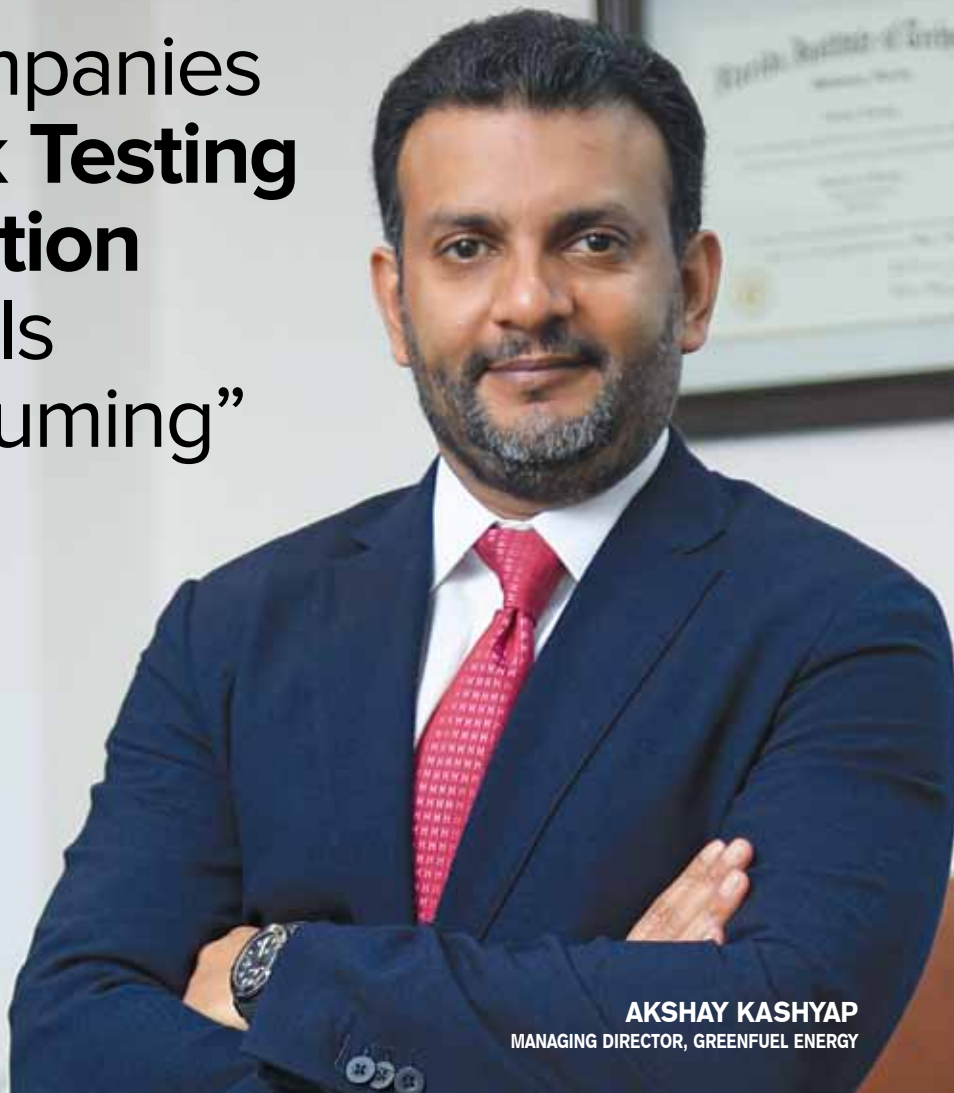




Industry

# “Many Companies Avoid **Pack Testing** And **Validation** Because It Is Time-Consuming”

Battery pack reliability goes beyond cell quality—it relies on precise engineering, thorough testing, and tailored thermal management. In a conversation with EFY’s Nitisha Dubey, Akshay Kashyap of Greenfuel Energy discusses whether the industry will prioritise safety and innovation or compromise quality for cost.

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## **How does proper engineering impact battery pack performance and cycle life compared to cell ratings?**

A battery pack’s performance heavily depends on its engineering, even if the cell is rated for 1000 cycles. Poorly designed packs fail to deliver the promised cycle life. Despite this, 90% of Indian manufacturers report cycle life based solely on cell data, avoiding time-intensive pack testing and

validation, which can take five to six months and incur high costs. As a result, reported figures often neglect the reduction in cycle life during the cell-to-pack transition. Depending on design and integration, cycle life varies widely, from 800 to 4000 cycles, with lifespans ranging between two and seven years.

## **How does an engineering-driven approach differ from basic assembly in battery pack manufacturing?**

An engineering-driven approach prioritises detailed analysis, such as computational fluid dynamics (CFD), to assess heat generation during charge and discharge cycles. This ensures the selection of optimal thermal materials, boosting safety and performance. In contrast, assemblers typically lack such in-depth analysis, resulting in substandard and potentially hazardous battery packs.